

The Circular Compartment Train (CCT): A Novel Sub-Linear Geometric Data Structure for Unique Set Topologies

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Abstract

This paper presents the *Circular Compartment Train (CCT)*, a highly adaptive, sub-linear data structure tailored for dynamic numeric sets. By blending a cyclically double-linked pointer system (the “train”) with a dynamic centralized geometric partitioning mechanism (the “compartments”), the CCT solves the linear lookup vulnerabilities inherent in standard linked layouts. We present two design topologies: the *CCT-Bag* for high-frequency multiset data streams and the *CCT-Set*, which leverages an immediate $\mathcal{O}(1)$ bitmap exclusion layout to prevent structural duplicate inflation. We prove that the balance bound governed by $M = \sqrt{N} + (0.05 \cdot N)$ preserves stable $\mathcal{O}(\sqrt{N})$ boundaries via structural compartment splits and flexible boundary shifts.

1 Introduction

Linear scanning limits standard linked list variants to an unfavorable $\mathcal{O}(N)$ lookup boundary. While structures such as Skip Lists or self-balancing Binary Search Trees (BST) bound operations to logarithmic intervals, they suffer from considerable structural pointer ballast and strict top-down layout constraints. We propose the Circular Compartment Train (CCT), an expressive hybrid topology that frames data elements into a rolling, continuous railroad ring, managed natively via an analytical multi-sector coordinate index representing radial pie slices mapping value intervals.

2 Structural Topology & Component Architecture

The data structure partitions architectural responsibilities across two structural segments:

1. **The Centralized Geometric Index:** A dynamic registry that partitions the total numeric domain into distinct intervals called *Compartments*. Each compartment maintains bounds [low, high], a size tracker, and a static entry pointer targeting the exact boundary element in the node ring.
2. **The Cyclic Train System:** A continuous, closed loop composed of doubly linked elements (*Wagons*), natively permitting bidirectionality (next and prev pointers). The physical terminus of the sequence loops back seamlessly into the initial node element, eliminating hard structural boundary limits.

3 Mathematical Balancing Operations

To maintain the runtime balance against targeted value clusters, individual compartment sizes are bounded by a dynamic equilibrium function M . The maximal wagon count per slice is tightly coupled to the square root of the system volume N enriched by an adaptive elasticity cushion:

$$M = \sqrt{N} + (0.05 \cdot N) \quad (1)$$

When an ingestion node hits the threshold limit where $|C_i| > M$, the CCT framework initializes a structural splitting routine:

$$\text{Threshold}_{\text{mid}} = \left\lfloor \frac{\text{low} + \text{high}}{2} \right\rfloor \quad (2)$$

The bounds are recalculated instantly, shifting the boundary definitions along the ring without forcing physical element allocation moves in memory.

4 Ingestion Filtering & Duplicate Invalidation

The *CCT-Set* model optimizes membership operations via a bit-vector signature mask assigned to each radial compartment slice. Before traversing the cyclic node network, incoming items undergo immediate indexing lookup:

$$f(X) = \text{Bitmap}[X \pmod{K}] \quad (3)$$

If the coordinate filter returns an active state, the system flags a collisions threat and rejects the element instantly. This short-circuit strategy bounds duplicate rejection tasks to a deterministic time complexity of $\mathcal{O}(1)$.

5 Complexity Overview

A systematic comparison underlines the competitive profile of the Circular Compartment Train against classical layouts under uniform distributions:

Table 1: Algorithmic Comparison Summary

Operation	Standard Linked List	Balanced BST	CCT (Ours)
Search Time	$\mathcal{O}(N)$	$\mathcal{O}(\log N)$	$\mathcal{O}(\sqrt{N})$
Insertion Time	$\mathcal{O}(1)$ or $\mathcal{O}(N)$	$\mathcal{O}(\log N)$	$\mathcal{O}(\sqrt{N})$
Duplicate Rejection	$\mathcal{O}(N)$	$\mathcal{O}(\log N)$	$\mathcal{O}(1)$
Memory Overlap Space	Moderate	High	Low-to-Moderate

6 Conclusion

The CCT architecture offers a compelling paradigm shift in layout design, blending cyclic connectivity with radial space splitting. Future milestones include the evaluation of dynamic slice merges under extreme deletion load states.